

Curriculum Vitae

Sangseok Han

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EDUCATION

University of Strathclyde — MSc, Advanced Naval Architecture

Degree awarded with *Distinction* (75.9/100, Highest MSc Classification)

United Kingdom
2024–2025

Inha University — BSc, Naval Architecture & Ocean Engineering

GPA: 3.7/4.0 (Rank: 2/16; First Class Equivalent)

South Korea
2021–2024

RESEARCH INTERESTS

- *Computational fluid dynamics (CFD)*
- *Near-wall turbulence and roughness effects*
- *Wave-structure interaction*

PUBLICATIONS

Peer-Reviewed Journal Articles

1. ***Beyond uniform roughness: Ship resistance with spatially non-uniform hull surface conditions***
Han, S., Lee, H., Dai, S., Terziev, M. (2026) — Ocean Engineering — Published; Corresponding Author.*
2. ***Temperature-induced errors in ITTC model-ship extrapolation — Cover Article***
Han, S., Dai, S., Terziev, M., et al. (2025) — Journal of Marine Science and Engineering — Published.
3. ***Resistance and speed penalty of a naval ship with hull roughness***
Choi, W., Min, G., Han, S., et al. (2024) — Ocean Engineering — Published.
4. ***How well do deterministic multi-component waves reproduce FOWT responses in JONSWAP seas?***
Han, S. (2026) — Applied Ocean Research — Under review; Solo Author.

Conference Papers

1. ***A Computational Fluid Dynamics Analysis of Spatial Roughness Distributions on Ship Resistance***
Han, S., Terziev, M., Tezdogan, T. (2026) — 36th Symposium on Naval Hydrodynamics (SNH 36) — Under review;*

RESEARCH EXPERIENCE

- **CFD Analysis of Ship Performance under Spatially Non-Uniform Hull Roughness**
University of Strathclyde (United Kingdom) (Oct 2024–Present)
— *Ocean Engineering — Published (2026); 36th Symposium on Naval Hydrodynamics — Invited full paper*
- **Deterministic and Stochastic Irregular Waves for Semi-Submersible FOWTs: A CFD-Based Analysis**
University of Strathclyde / The University of Osaka (Dual-affiliated project) (July 2025–Present)
Submitted to Applied Ocean Research — Under review.
- **Simulation-Based Modelling and Control of Wave Energy Converters (WECs)**
Visiting Researcher, The University of Osaka (Japan) (Oct 2025–Nov 2025)
— *Presented at JASNAOE 2025*
- **Numerical Investigation of Fluid-Structure Interaction in Semi-Submersible FOWT**
University of Strathclyde (United Kingdom) (Sep 2024–Sep 2025)
— *Presented at MARINE 2025.*
- **CFD Study on the Temperature-Sensitivity of the ITTC 1957 Correlation Line**
Inha University (South Korea) (Jan 2024–June 2025)
— *Published as Cover Article in JMSE (2025); Featured in University News (2025).*

CONFERENCE PRESENTATIONS

Selected Conferences

* Corresponding author for all works listed below.

- **36th Symposium on Naval Hydrodynamics (Full Paper Invited, ONR), 2026 (Under review).**
A Computational Fluid Dynamics Analysis of Spatial Roughness Distributions on Ship Resistance
- **XI International Conference on Computational Methods in Marine Engineering (MARINE), 2025.**
Numerical Analysis of Fluid-Structure Interaction (FSI) for Semi-Submersible Floating Offshore Wind Turbine
- **64th International Congress of Naval Architecture, Marine Technology and Maritime Industry, 2025.**
Impact of Towing Tank Temperature on Model-Ship Extrapolation: Revisiting the ITTC Procedure
- **The Japan Society of Naval Architects and Ocean Engineers (JASNAOE) Autumn Meeting, 2025.**
CFD-Based Model Predictive Control for Point Absorber Wave Energy Converters
- **13th National Congress on Fluids Engineering (Best Paper Award), 2024.**
Uncertainty of the ITTC 1957 Correlation Line with Temperature Variations

HONOURS AND AWARDS

- **Cover Article (1 out of 185) | *Journal of Marine Science and Engineering* (July 2025).**
- **Best Paper Award | The 13th National Congress on Fluid Engineering (July 2024).**
- **Reviewer Excellence Certificate | *Ocean Engineering* (Q1, IF 5.5) (Feb 2025).**
- **Dean's International Excellence Award | University of Strathclyde (June 2024).**
- **Featured in the Official News | University of Strathclyde (Aug 2025)**
- **International Conference Travel Bursary | The Worshipful Company of Shipwrights (Apr 2025).**
- **Industrial Technology Award | The Society of Naval Architecture of Korea President (Aug 2023).**
- **Autonomous System Award | GNTP Chief (August 2022, August 2023).**
- **Ship Design Contest Excellence Award | Korea Offshore & Shipbuilding Association (Nov 2022).**
- **Academic Excellence Scholarship | Inha University (January 2024, Aug 2022).**
- **Merit-Based Scholarship | Inha University (June 2024).**

ACTIVITIES

Academic Engagement

- **Special Research Student (Visiting Researcher), The University of Osaka (Oct 2025 – Dec 2025)**
- **Peer Reviewer, *Ocean Engineering* (2025 – Present)**
↳ Award: Reviewer Excellence Certificate (2025).
- **Peer Reviewer, *Physics of Fluids* (2025 – Present)**
- **Registered Reviewer, *Applied Ocean Research* (2025 – Present)**
- **BSc Capstone: FOWT and Wave-breaker Interaction Analysis Using CFD and EFD (2023)**
- **Student Ambassador, *The Korean Society of Ocean Engineering* (2022–2023)**
- **Ship Design Contest: Concept Design of a 20,000 CBM Nuclear Fusion Propulsion LH2 Carrier (2022)**

Leadership and Technical Roles

- **Algorithm Leader, Autonomous Boat Development (2021–2024)**
Marine Robotics Society (*University of Strathclyde*), i-Tricat (*Inha University*)
↳ Developed autonomous navigation algorithms for robotic surface vessels.
- **Mathematics Tutor (2021–2024)**
- **Military Service, Republic of Korea Navy (2016–2020)**
Petty Officer, operated sonar systems for underwater detection.
↳ Awards: Commander of the 2nd Fleet Commendation (2020), Captain's Commendation Medal (2018).